EEMBC-style PQC benchmarking on a modern Cortex-M33

SecureMark-V2 extended with native post-quantum primitives — progress report for Prof. Tschofenig

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Executive summary

The question this work answers is whether post-quantum cryptography can be benchmarked inside a standard, recognised harness on a modern microcontroller, and what it costs relative to classical cryptography on the same silicon. It can, and the answer is more favourable than expected.

The official EEMBC SecureMark-V2 harness now runs 102 components on the FRDM-MCXN947 — 82 classical plus 20 native post-quantum — and emits its own score sheet. All 20 PQC components pass their functional self-check and their regression checksum. The post-quantum public-key phase (ML-KEM-768 + ML-DSA-65) costs **99.5 ms against 94.9 ms for classical P-256**, a penalty of **4.9%**.

The four numbers that matter

Measure	Result	What it means
PQC_All score	**2042.526**	20 per-primitive PQC components, equal weight, harness-computed
PQC vs classical handshake	**+4.9%** (99.5 vs 94.9 ms)	Post-quantum is a few percent dearer, not multiples
Library spread at matched -O2	**2.41x** (wolfSSL 6.37 vs PQClean 15.33 ms)	Implementation choice matters more than the algorithm choice
128 KB firmware verification	HashML-DSA-65 **24.45 ms** vs ECDSA **26.70 ms**	PQC verification is already faster than the classical baseline

> One result deserves emphasis because it inverts the usual expectation: at -O2 with a modern engine, **the post-quantum handshake is within 5% of classical**, and PQC signature verification is faster than ECDSA. The cost sits in signing, and in the size of what has to be carried.

1. What was built

Component	Role
`wolfssl-m33/`	crypto-only wolfSSL 5.9.2 cross-built for Cortex-M33 with ML-KEM and ML-DSA enabled; static pool allocator (no heap); deterministic RNG for reproducible checksums
`bench/`	standalone driver: 7 algorithms x keypair / sign-or-encapsulate / verify = 21 measurement points
`v2harness/`	upstream EEMBC securemark-v2 self-hosted harness, extended with 20 native PQC components registered as the PQC_All suite
`task3a/`	128 KB firmware signature verification (the secure-boot case)
`results/PQC_LIBRARY_CONFIG_MATRIX.csv`	42-row library and configuration matrix

Platform port

The harness expects an operating system. Running it bare metal required a thin porting layer: the DWT cycle counter replaces `clock_gettime` (64-bit accumulated, so the 32-bit CYCCNT wrap cannot corrupt a measurement), `_write` is retargeted to the LPUART, and both `th_malloc` and wolfSSL's `XMALLOC` are routed to a single 160 KB static pool so allocation behaviour is deterministic.

Three harness-level defects had to be fixed before any run could complete. They are listed in section 5 because they are methodologically relevant, not incidental.

2. Correctness first: the standalone bench

Before trusting a harness, the primitives have to be shown to run correctly on the target and to produce identical output across builds. Seven algorithms, 21 points, all passing:

Algorithm	keypair	encaps / sign	decaps / verify	Check	Regression CRC
ML-KEM-512	17.43	14.97	20.24	PASS	0xBB6188B7
ML-KEM-768	28.02	25.24	32.74	PASS	0x8A48D4A5
ML-KEM-1024	44.10	40.06	49.84	PASS	0xFDB3B688
ML-DSA-44	66.36	165.55	69.70	PASS	0xBA561067
ML-DSA-65	113.20	198.77	115.62	PASS	0x1EE8F069
ML-DSA-87	192.68	254.21	195.14	PASS	0xAB6329B9
HashML-DSA-65	113.20	235.17	115.11	PASS	0x9C084B80

Milliseconds, DWT cycles ÷ 150 MHz, -Os first pass (the -O2 figures follow).

On the checksums. EEMBC has published no PQC profile, so there is no golden CRC to match against for ML-KEM or ML-DSA. Those components are therefore reported under a dual scheme and are never labelled “EEMBC CRC MATCH”: a functional round-trip self-check (KEM decapsulation recovers the encapsulated secret exactly; DSA verification returns valid) plus a synthesised regression CRC over the deterministic output. The CRC values reproduce across independent builds and across optimisation levels, which is what makes them usable as a cross-build invariant rather than a build artefact. The classical suites remain the only ones checked against EEMBC goldens.

3. The native score sheet

Emitted by the harness itself — not reconstructed from a serial capture. 102 components, zero CRC errors.

Attribution. The harness framework, the three suite definitions and their golden CRCs are EEMBC's (upstream `eembc/securemark-v2`). The 20 post-quantum components, their `th_` shims and the `PQC_All` suite are **ours**, added on top. **PQC_All and PQC_TLS1.3 are therefore not EEMBC scores** — the harness prints its own disclaimer that these are not official. Only the classical suite scores are EEMBC-comparable.

Suite	Score	Suite	Score
PQC_All	**2042.526**	Boot_Med_RSA	38.160
PQC_TLS1.3 (768/65)	**40177.367**	Boot_Med_ECC	24.572
TLSv1.2	10418.526	TLSv1.3_Heavy_CCM	4.127
TLSv1.3_Light	251.907	TLSv1.3_Heavy_GCM	4.124
Boot_Light	70.419	Boot_Heavy_RSA	10.408
TLSv1.3_Med_CCM	100.576	Boot_Heavy_ECC	1.178
TLSv1.3_Med_GCM	100.162		

Score = $1000 / \Sigma(w_i / ips_i)$. PQC_All uses w = 1 over the 20 per-primitive components; PQC_TLS1.3 uses w = 0.25 over ML-KEM-768 encapsulate/decapsulate and ML-DSA-65 sign/verify.

PQC components at -O2 (30-iteration means)

Algorithm	keypair	encaps / sign	decaps / verify
ML-KEM-512	4.36	3.76	5.08
ML-KEM-768	7.01	6.37	8.24
ML-KEM-1024	11.07	10.14	12.56
ML-DSA-44	14.50	29.18	14.53
ML-DSA-65	25.13	60.36	24.57

ML-DSA-87	43.24	80.22	42.42
HashML-DSA-65	(as ML-DSA-65)	55.68	24.45

Milliseconds. All 20 components: self-check PASS, regression CRC matched.

4. Where the cost actually is

4.1 PQC versus classical, same library, same flags

Both sides wolfSSL at -O2. Public-key phase = one ECDH + one signature + two verifications:

Public-key phase	Composition (ms)	Total
Classical P-256	24.34 ECDH + 17.57 sign + 2 × 26.50 verify	**94.91 ms**
PQC (ML-KEM-768 + ML-DSA-65)	6.37 + 8.24 + 60.36 + 24.57	**99.54 ms**

A **4.9% penalty**. Post-quantum key establishment is far cheaper than its classical counterpart (6.4 + 8.2 ms versus 24.3 + 26.5 ms); the whole penalty, and more, is carried by ML-DSA signing at 60.4 ms.

4.2 Library comparison, controlled at one optimisation level

ML-KEM-768 encapsulation	Implementation	Flags	Time	Cycles
Reference	PQClean clean C	-O2	15.33 ms	2,299,500
Embedded engine	**wolfSSL native**	-O2	**6.37 ms**	**955,000**
Hand-tuned assembly	pqm4 (Cortex-M4F asm)	-O3	23.99 ms	3,598,500

Two results worth reporting plainly. **wolfSSL's portable C is 2.41x faster than the reference clean-C implementation** at identical flags — a library effect, not a compiler effect. And **the hand-tuned Cortex-M4F assembly is the slowest of the three**: it is not portable to ARMv8-M, its decapsulation and all of ML-DSA fault on the M33, and its encapsulation is 1.6x slower than plain clean C at matched -O2. The expected ordering — reference, then portable engine, then assembly — does not hold on this core.

ML-DSA-65 signing shows the same pattern: PQClean 91.61 ms mean at -O2 against wolfSSL 60.36 ms, a **1.52x** library effect. Optimisation level is almost irrelevant to PQClean's ML-DSA (median 73.72 ms at -Os versus 74.77 ms at -O2, a 1.01x change), so the gap is the library, not the compiler.

4.3 Optimisation sensitivity

Same code, same board:

Quantity	-Os	-O2	Factor
PQC_All	414.746	2042.526	4.9x
PQC_TLS1.3	8525.134	40177.367	4.7x
wolfSSL ML-KEM-768 encapsulate	25.21 ms	6.37 ms	4.0x
PQClean ML-KEM-768 encapsulate	37.9 ms (-O0)	15.33 ms	2.47x
PQClean ML-DSA-65 sign (median)	73.72 ms	74.77 ms	1.01x

This is why the optimisation level had to be fixed before any comparison: a five-fold difference across the board, and near-zero for one primitive. **-O2 is now the stated level for every number in this report.**

5. The application benchmarks

5.1 Task 3A — 128 KB firmware signing (secure boot)

Payload 131,072 B, 20 repeats per scheme, verification is the decision a bootloader actually makes:

Scheme	sign (mean / min / max)	**verify**	sig bytes	payload RAM
ECDSA P-256	16.93 / 16.93 / 16.93	**26.70 ms**	72	streamed
ML-DSA-44 pure	155.72 / 137.42 / 217.14	**133.10 ms**	2420	**131,072 B**
ML-DSA-65 pure	179.47 / 148.75 / 318.18	**143.13 ms**	3309	**131,072 B**
HashML-DSA-65	55.30 / 30.08 / 107.91	**24.45 ms**	3309	streamed

Three findings. **Hashing dominates pure ML-DSA verification** — the 143 ms figure is mostly a SHAKE256 pass over 128 KB inside the primitive; pre-hashing lifts it out and verification drops to 24.45 ms, 5.9x faster. **HashML-DSA verification beats ECDSA P-256 verification** (24.45 versus 26.70 ms). And the buffering split is real: pure ML-DSA needs the whole payload addressable at once, so a device receiving firmware over a link must hold 128 KB of RAM, whereas ECDSA and HashML-DSA stream in 4 KB chunks with a 112 or 224 B hash context.

5.2 Task 3B — 256 B telemetry signing and the fat tail

Level	median	p99	max	worst / median
ML-DSA-44	43.94 ms	181.91 ms	289.05 ms	6.6x
ML-DSA-65	73.72 ms	293.99 ms	505.68 ms	6.9x
ML-DSA-87	99.41 ms	336.24 ms	477.86 ms	4.8x

n = 2000 per level. At a 10 Hz telemetry budget (100 ms/frame) the median fits at every level and the worst case does not. ML-DSA signing is rejection-sampled, so it has a fat right tail; a design that checks only the median is not safe.

6. Method notes

These matter because they are the difference between a number and a defensible number.

Practice	Why
Frozen split, single test touch	The conformal calibration uses a frozen four-way split; the test set is touched exactly once, with every threshold pre-frozen. An earlier contaminated protocol produced a flattering result that did not survive this discipline, and it was withdrawn.
Dual evidence for PQC, goldens for classical	No EEMBC PQC profile exists. PQC components report a functional round-trip check plus a synthesised regression CRC; only the classical suites claim EEMBC CRC MATCH.
One stated optimisation level	A five-fold spread across -Os/-O2 made mixed-flag comparisons meaningless. All numbers here are -O2.
Means, not single draws	ML-DSA signing is rejection-sampled. Harness PQC components run 30 local iterations; distributions come from 2000-run measurements.

Three harness defects were found and fixed in the course of this work, each of which had been silently preventing a complete run: the classical P-384 ECDH shim passed a null RNG (`MISSING_RNG_E`); RSA-4096 was outside `RSA_MAX_SIZE` because `SP_INT_BITS` defaulted to 2048; and the autotune loop doubled iterations until one second of work elapsed, so a failing component spun forever and the score section never printed. All three are documented at the point of change.

7. Status against the brief

Reported explicitly, because the gaps are the useful part of a progress report.

Task	Requirement	Status
1	SecureMark-V2 with native ML-KEM / ML-DSA interfaces	**Done**
1	HashML-DSA (pre-hash) pipeline	**Done** — measured separately from pure ML-DSA
1	Explicit ML-DSA context strings	Open — shims call ctx = NULL, 0
1	Realistic message lengths	Partly — 128 KB firmware and 256 B telemetry done; 4 KB certificates not
2	Library / configuration matrix	**Done** — 42 rows; PQClean, wolfSSL, mbedTLS-classical, pqm4
2	mbedTLS-PQC and mldsa-native variants	Not included
3A	128 KB firmware verification, PQC vs classical	**Done**
3B	256 B telemetry fat tail, 2000 runs	**Done**
3C	TLS / DTLS 1.3 with RFC 9146 Connection IDs	Open — existing CID work predates V2 and is a model, not a measurement
4	Automation: evaluation script and summary CSV	**Done**
4	LaTeX table generation	Open
4	Per-primitive peak stack	Open in the harness — no stack painting in the PQC components
Acc.	Clean build under Zephyr v4.2.0-rc1 (MCXN947)	**Open**
Acc.	Clean build under ESP-IDF v5.1 (ESP32-S3)	**Open**
Acc.	CRC MATCH	Classical 14/14 against EEMBC goldens; PQC self-check + regression CRC
Acc.	SECUREMARK_V2_PQC_REPORT.md	**Done**

8. What remains, in order

The single largest gap is that everything here is one platform. The Zephyr and ESP-IDF ports named in the acceptance criteria are not built. They are build-system work on code that already compiles and runs on the M33, not new measurement, but until they exist the result is single-platform. After that: LaTeX emission from the existing CSV, the ML-DSA context-string parameter, per-op stack painting, and Task 3C.

Everything reported above is reproducible from the artifacts in the appendix, and every table can be regenerated from the raw captures.

Appendix — artifacts

Path (under `codebase/projects/securemark_v2_pqc/`)	Contents
`v2harness/results/SECUREMARK_V2_FINAL_REPORT.md`	full report with the complete correction trail
`v2harness/results/pqc_denoised.txt`	raw serial capture of the score-sheet run
`v2harness/raw/SECUREMARK_V2_PQC_SUMMARY.csv`	per-component ips and CRCs
`results/PQC_LIBRARY_CONFIG_MATRIX.csv`	library matrix, 42 rows
`bench/results/STANDALONE_PQC_BENCH.md`	standalone bench
`task3a/results/task3a_dist.txt`	128 KB firmware verification, n = 20
`mldsa_bench_nxp/results/task3b/`	2000-run signing distributions
`scripts/evaluate_securemark_v2_pqc.py`	regenerates the scores from a capture